

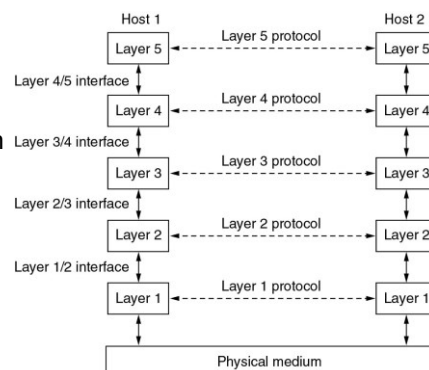
01 Introduction

01.02-01.03 OSI/ISO and TCP/IP models, network examples

Copyright Note: all figures and tables reported, if not specified, are taken from: A. Tanenbaum: *Computer Networks*, 4th ed. Pearson Prentice Hall
(Others are expressly specified)

Layered software organization

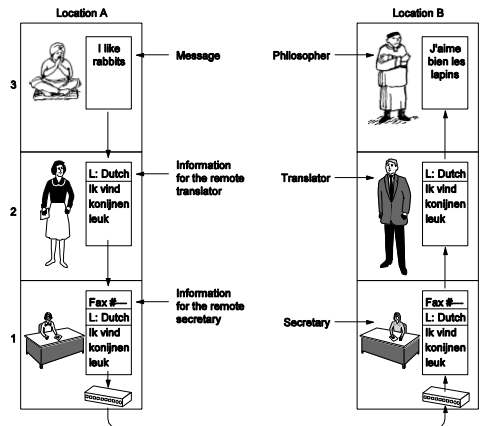
- Network software needs to be structured in layers to **handle complexity**
- Each layer gives **services to its upper layer** hiding internal details
- **Layers can be seen as virtual machines** for the layers that are on the upper levels
- **Peer layers communicate by means of a protocol**
- Contiguous layers are connected by **interfaces** that define the services of the lower layer for the upper layer
- Layers + protocols = Network architecture



Layers protocols and interfaces

Protocol Hierarchies

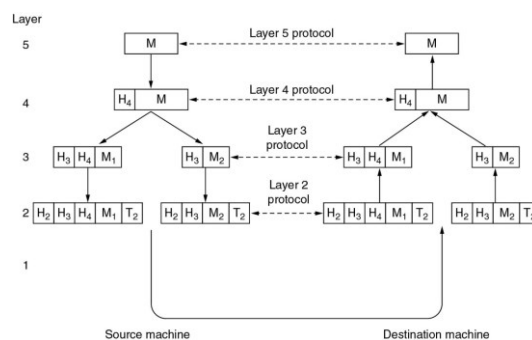
- An example of layered architecture



The philosopher-translator-secretary architecture.

Protocol Hierarchies

- Relationships between virtual (horizontal) and real (vertical) communication
- Peer process abstraction as fundamentals network design principle
- Lower layers may be implemented in hardware



Example information flow supporting virtual communication in layer 5.

Design Issues for the Layers

- Addressing
 - To identify source and destinations
- Error Control
 - To take into account the effect of real world physical links
- Flow Control
 - To avoid the overloading of a slow receiver by a high-speed source
- Multiplexing
 - To allow possible multiple independent communications on the same connection
- Routing
 - To decide the path from source to destination

Services classification

- **Connection-oriented**
 - Connections are established, used and released
 - Connection parameters may be negotiated
- **Connectionless**
 - Each message is independently sent along the system independently from the others
 - Each message needs to know source and destination
- **Quality of service classification:**
 - Reliable: data are not lost (usually based on **acknowledgments**)
 - Unreliable: data may be lost (also called **datagram services**)

Services classification

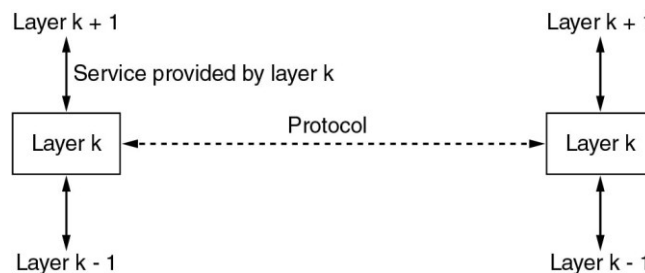
- Some example of different services

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Remote login
	Unreliable connection	Digitized voice
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Registered mail
	Request-reply	Database query

Example of six different types of service.

Services to Protocols Relationship

- Protocols and services should not be confused
- Services are collections of primitive operations offered by a layer to its upper layer
- Protocols are sets of rules that specify the format and the semantic of messages (packets) exchanged between peer entities of the same layer



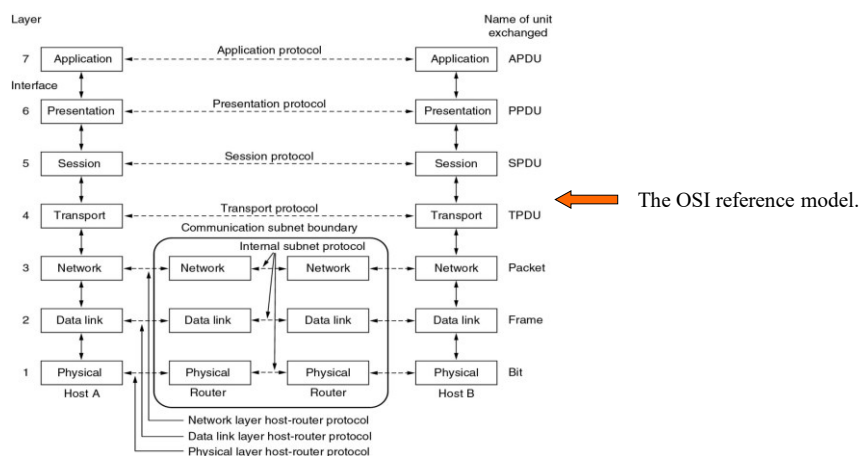
The relationship between a service and a protocol

Reference Models

- Analysis of two reference models of protocol stacks and compare them
- The OSI Reference Model (standard de iure)
- The TCP/IP Reference Model (standard de facto)
- Comparison between OSI and TCP/IP models

Reference Models: ISO OSI

- Not used in practice but useful as a reference model (standard de iure)
- 7 layers from physical layer up to application layer



Physical Layer

- Media where are sent bits between nodes
 - Physical interfaces and media
 - Bit representation
 - Speed transmission
 - Synchronization
 - Link
 - Topology
 - Data flow

Data Link Layer

- Reliability of data transmission
 - Framing
 - Physical addressing
 - Flow control
 - Error control
 - Access control

Network Layer

- Network layer deliver data packets among nodes of a net.
 - Logic addressing
 - routing

Transport layer

- Delivers messages among sender processes and receiver processes
 - Process addressing
 - Segmentation
 - Connection control
 - Flow control
 - Error control

Session layer

- Does synchronization and dialog control
 - Synchronization
 - Dialog control

Presentation layer

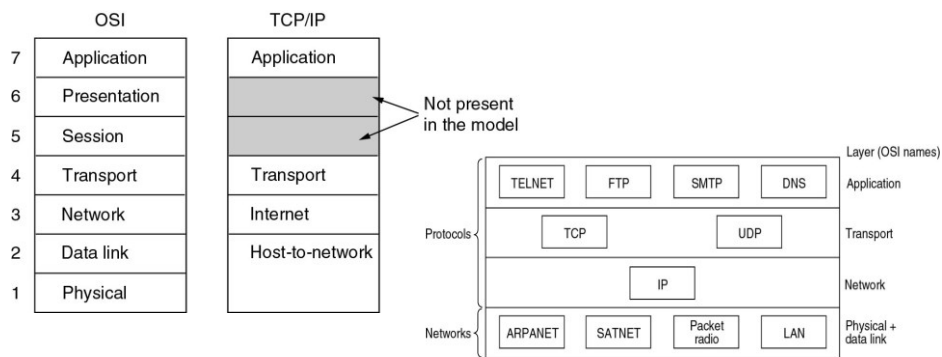
- Translation
- Ciphering
- compression

Application layer

- Give network services to final user
 - Virtual terminal
 - File manager
 - Mail
 - Network directory

Reference Models: TCP/IP

- The TCP/IP reference model (left) and some relationship between TCP, IP, application protocols and networks (right)



The TCP/IP reference model

Protocols and networks in the TCP/IP model initially

Comparing OSI and TCP/IP Models

- Concepts central to the OSI model
 - Services, Interfaces and Protocols are well defined and separated but not linked to any real network
- Concepts central to the TCP/IP model
 - Protocols were born before the model description, so they are perfectly coherent to the model. The IP protocol allows flexible packets exchange along a great variety of networks

Comparing OSI and TCP/IP models

- OSI:
 - Problems: too much complex, scarce implementations, (vs. Unix TCP/IP implementations)
- TCP/IP:
 - Problems: not a general model, Host-to-network “layer” is not really a layer, no mention of physical and data link layers

Proposal: a hybrid model

- A good way of reasoning about networks is to try to take some of the best from both TCP/IP and OSI in a hybrid model:
 - The protocol stacks model conceptually based on OSI
 - The widely used TCP/IP family protocols

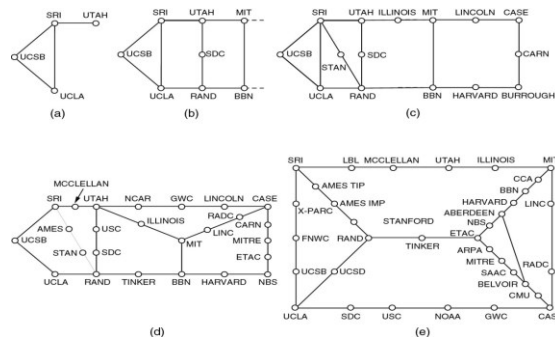
5	Application layer
4	Transport layer
3	Network layer
2	Data link layer
1	Physical layer

Example Networks

- The Internet
- Connection-Oriented Networks:
X.25, Frame Relay, and ATM
- Ethernet
- Wireless LANs: 802.11

The ARPANET

- ARPANET was born as a project of ARPA (Advanced Research Project Agency)
- This project stimulated research in protocols and networks → TCP/IP (1974)
- Protocols became integrated in Berkeley Unix
- Lots of universities started to connect their computer in LANs and then LANs to the ARPANET
- In late 80's the National Science Foundation (NSF) started developing a network for academic purposes to connect each other research group autonomously from the Defense Department
- The interconnection of ARPANET and NSFNET working under the TCP/IP protocols has been the mother of the current Internet

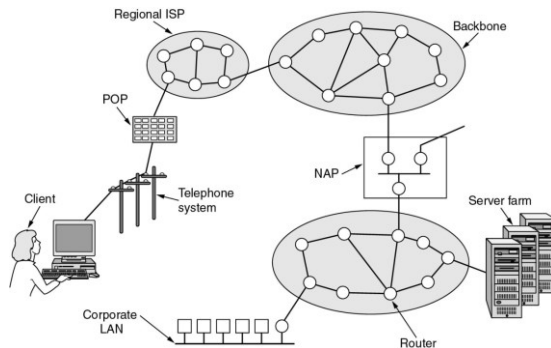


Growth of the ARPANET (a) December 1969. (b) July 1970(c) March 1971. (d) April 1972. (e) September 1972

Internet Usage

- Traditional applications (1970 – 1990) :
 - E-mail
 - News
 - Remote login
 - File transfer
- Up to 1990 mainly academic or military applications, but then...
 - World Wide Web: users exponential growth
 - Internet Service Providers, .com and other new economy companies birth

Architecture of the Internet



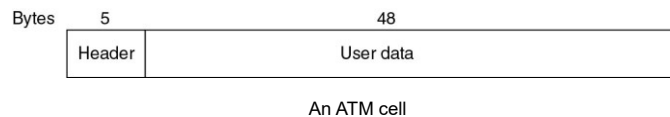
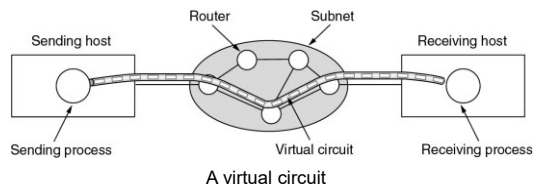
- A typical overview of the Internet
- Legend:
 - POP: Point of Presence
 - ISP: Internet Service Provider
 - NAP: Network Access Point

Connection oriented networks

- The datagram model (connectionless) is typical of subnets whose main target was faults tolerance (if some routers crash and if there are alternative paths, packets may follow a different path from those preceding the crash)
- Connection oriented model is typical of telephone companies. When the communication is established, packets follow the same path. Why is preferred by these type of companies?
 - Quality of Service
 - Billing
- Some example of connection-oriented networks:
 - X.25 (first public data network in '70)
 - Frame relay (after X.25, still used in some case for connecting LANs in companies)
 - ATM: developed by telecom carriers (widely used in telephone networks to carry IP packets)

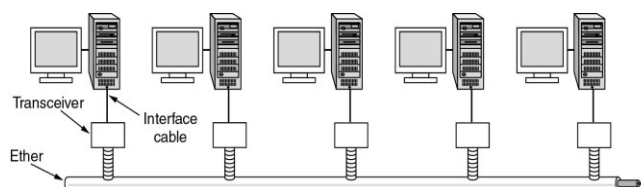
ATM Virtual Circuits

- ATM (Asynchronous Transfer Mode) is based on the concept of virtual circuits
- Data is sent in the form of small packets called cells (5+48=53 bytes)
- The header contains the information regarding the connection
- Reduced cells dimension allows easy construction of hardware routers
- Some typical bit rate for ATM networks: 155 Mbps and 622 Mbps



Ethernet

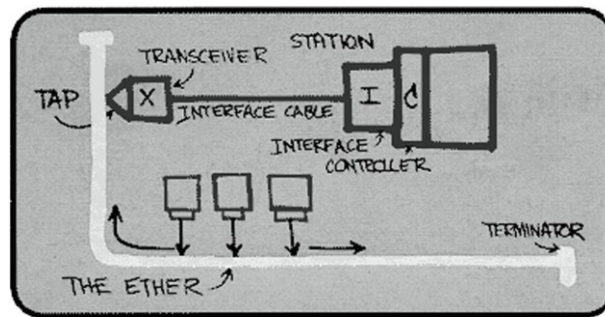
- The first project (Metcalfe and Boggs, 1976) was able to connect up to 256 computers on a multidrop coaxial cable at 2.94 Mbps and was called Ethernet Xerox
- In 1978 DEC, Intel and Xerox developed the 10 Mbps version of the system which became, due to its success, the IEEE 802.3 standard
- Ethernet is not the only standard for LANs: 802.4 (token bus) and 802.5 (token ring) are other defined standard but they are not currently used



Architecture of the original Ethernet

Ethernet

- Some network archaeology (30 years ago)...

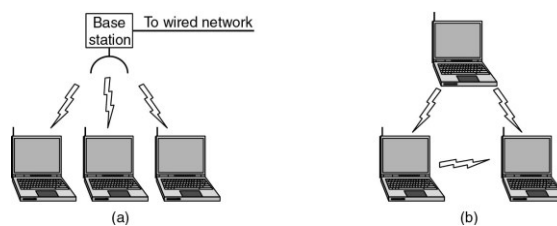


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♦ Metcalfe's original conception of the Ethernet

Wireless LANs

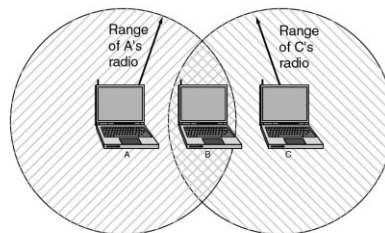
- The wireless standard 802.11 (also known as WiFi) was created to support a wireless version of the cabled LANs
- 2 operating modes
 - Base station
 - Ad hoc networking
- 802.11 is compatible with Ethernet on layers upper to the data link layer
- But there are several issues in physical and data link layers that the standard must take into account (see next 2 slides)



(a) Wireless networking with a base station and (b) Ad hoc networking

Wireless LANs (2)

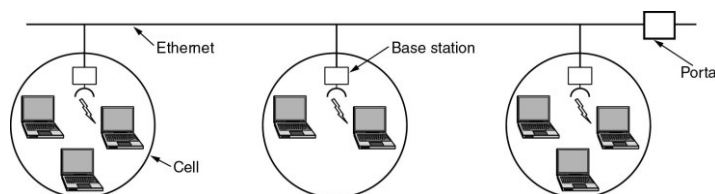
- In general in Ethernet, each hosts listens to the channel and is allowed to transmit if the channel is free
- This cannot be done in wireless (see figure)
 - If C wants to transmit to B it cannot detect if also A is transmitting something to B



The range of a single radio may not cover the entire system.

Wireless LANs (3)

- Other typical problems to be faced are those coming from the transit of a computer from area X covered by base station x to area Y covered by base station y
- The network reported below is a possible solution for handling the mobility. It can be seen as an Ethernet by the 'rest of the world'



A multicell 802.11 network.

IEEE 802 Standards

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)
802.13	Unlucky number. Nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 *	Personal area networks (Bluetooth)
802.16 *	Broadband wireless
802.17	Resilient packet ring

The 802 working groups. The important ones are marked with *. The ones marked with ↓ are hibernating.

Metric Units

Exp.	Explicit	Prefix	Exp.	Explicit	Prefix
10 ⁻³	0.001	milli	10 ³	1,000	Kilo
10 ⁻⁶	0.000001	micro	10 ⁶	1,000,000	Mega
10 ⁻⁹	0.000000001	nano	10 ⁹	1,000,000,000	Giga
10 ⁻¹²	0.000000000001	pico	10 ¹²	1,000,000,000,000	Tera
10 ⁻¹⁵	0.000000000000001	femto	10 ¹⁵	1,000,000,000,000,000	Peta
10 ⁻¹⁸	0.000000000000000001	atto	10 ¹⁸	1,000,000,000,000,000,000	Exa
10 ⁻²¹	0.000000000000000000001	zepto	10 ²¹	1,000,000,000,000,000,000,000	Zetta
10 ⁻²⁴	0.00000000000000000000001	yocto	10 ²⁴	1,000,000,000,000,000,000,000,000	Yotta

- The principal metric prefixes